MACHINE LEARNING ENGINEERING LAB SERIES

Executive Diagnostic Brief

Five practical exercises covering regression, classification, ensemble learning, and customer segmentation.

Author	Hitesh Sharma
Project	ML Engineering Lab Series
Environment	Python 3.12, Scikit-learn, Pandas, NumPy, Matplotlib, Jupyter
Repository	github.com/Hitesh0825/ML-engineering-lab-series

Prepared from the project setup, lab acceptance criteria, and metric outputs shared during the workflow. Unverified metrics are explicitly marked as requiring confirmation.

1. Executive Summary

This project implements five machine-learning exercises: insurance cost regression, bank-marketing classification with Logistic Regression, Decision Tree classification, Random Forest classification, and K-Means customer segmentation. The notebooks and output CSV files were created and pushed to the project's GitHub repository.

The Logistic Regression exercise has verified metrics that exceed the specified acceptance thresholds. The K-Means exercise produces four clusters, but its current silhouette score is below the lab target. Metrics for Exercises 1, 3, and 4 should be confirmed from their final notebook outputs before claiming that all acceptance criteria passed.

2. Project Scope and Deliverables

Deliverable	Location	Status
Exercise notebooks	notebooks/01–05*.ipynb	Uploaded
Raw datasets	data/raw/	Uploaded
Prediction / cluster outputs	outputs/	Several CSVs confirmed in Git commit
Exercise-specific documentation	exercise-readmes/	Created; repository should be checked
Executive diagnostic brief	reports/	This PDF

3. Exercise Findings

Exercise 1 — Linear Regression: Insurance Cost Prediction

Predict medical insurance charges from customer attributes using a regression workflow. The notebook reportedly includes raw-target and log-target modeling, diagnostics, coefficient interpretation, and acceptance checks.

Acceptance criteria: Lab targets: $R^2 \geq 0.75$; RMSE < 4,800; estimated smoker effect > 23,000.

Result / status: Actual numeric results are not available in the shared evidence used for this brief. Confirm final notebook outputs before reporting pass/fail.

Exercise 2 — Logistic Regression: Bank Marketing

Binary classification of whether a bank customer subscribes to a term deposit. The workflow uses preprocessing for numeric and categorical features, class weighting, probability thresholds, and classification metrics.

Acceptance criteria: ROC-AUC target ≥ 0.78; recall target ≥ 0.70 at threshold 0.35.

Result / status: VERIFIED: ROC-AUC = 0.8769 (pass). Recall at threshold 0.35 = 0.8552 (pass). Prediction output: outputs/exercise2_predictions.csv.

Exercise 3 — Decision Tree Classification

Compare Gini and entropy criteria, constrain tree depth, inspect interpretable rules, and evaluate classification performance on bank-marketing data.

Acceptance criteria: Maximum depth ≤ 5; test accuracy ≥ 0.85.

Result / status: The notebook and prediction CSV were created. Final accuracy/depth output was not provided in the shared evidence; verify before marking the criteria as passed.

Exercise 4 — Random Forest Classification

Train a bootstrap Random Forest, evaluate classification metrics and out-of-bag performance, and inspect feature importance.

Acceptance criteria: OOB score ≥ 0.88; F1 score ≥ 0.65.

Result / status: Notebook, prediction CSV, and feature-importance CSV were created. Final OOB/F1 values were not provided in the shared evidence; verify before marking the criteria as passed.

Exercise 5 — K-Means Customer Segmentation

Segment wholesale customers using six spending features: Fresh, Milk, Grocery, Frozen, Detergents_Paper, and Delicassen. Preprocessing uses log1p transformation and scaling; K values from 2 through 8 were compared.

Acceptance criteria: Lab configuration K = 4; silhouette target ≥ 0.42.

Result / status: NOT MET: StandardScaler silhouette at K=4 = 0.1885. Best tested StandardScaler result was K=2 with 0.2903. RobustScaler at K=4 = 0.2017; Normalizer at K=4 = 0.2098. Keep actual values and limitation transparent. Outputs: outputs/exercise5_customer_clusters.csv and outputs/exercise5_cluster_profiles.csv.

4. Technical Approach

The project uses Python notebooks for reproducible exploration and modeling. Pandas and NumPy support data handling; Scikit-learn provides preprocessing, model training, and evaluation; Matplotlib/Seaborn support visual analysis. The bank-marketing CSV uses a semicolon delimiter and contains a 21-column UCI-style variant without the `balance` field found in some alternative versions. The implementation uses available fields rather than inventing a missing column.

5. Risks, Limitations, and Recommendations

• Confirm final metrics for Exercises 1, 3, and 4 from the executed notebooks.

• For Exercise 5, report that the K=4 silhouette target was not met; do not state that all clustering acceptance criteria passed.

• The bank-marketing dataset variant differs from some schemas. Document the actual fields used and retain the dataset source/version in project notes.

• Keep virtual environments such as .venv and .venv312 excluded from Git; share dependency specifications through requirements.txt instead.

• Check that the main README, all five exercise README files, notebooks, outputs, and this PDF are visible in the GitHub repository.

• If the lab requires every criterion to pass, revisit clustering feature selection/validation and record any justified model changes with their actual scores.

6. Repository and Reproducibility

Repository: https://github.com/Hitesh0825/ML-engineering-lab-series

Key folders: `notebooks/`, `data/raw/`, `outputs/`, `exercise-readmes/`, `reports/`, and `src/`. Use Python 3.12 as the intended environment, install dependencies from `requirements.txt`, and execute notebooks in numerical order. Generated results may vary if dependencies, random seeds, dataset versions, or preprocessing steps change.

Conclusion

The repository contains a complete five-exercise implementation scaffold with documented outputs. Logistic Regression has verified passing metrics; K-Means currently falls short of the silhouette target; and remaining exercise metrics need a final evidence-based check. The report intentionally distinguishes verified results from metrics that were not supplied.

End of report